

A Mini Project Report on
Digital Notary System

Submitted By

Samiksha Deo	21104124
Sushant Mhatre	21104002
Siddhi Desale	21104135
Siddharth Devare	21104136

Under The Guidance of
Mr. Mandar Ganjapurkar



DEPARTMENT OF INFORMATION TECHNOLOGY

A.P. SHAH INSTITUTE OF TECHNOLOGY
G.B. Road, Kasarvadavali, Thane (W), Mumbai-400615

UNIVERSITY OF MUMBAI

Academic Year: 2024-25

CERTIFICATE

This to certify that the Mini Project report on **Digital Notary System** has been submitted by Samiksha Deo(21104124), Sushant Mhatre (21104002), Siddhi Desale(21104135) and Siddharth Devare(21104135) who are a Bonafide students of A. P. Shah Institute of Technology, Thane, Mumbai, as a partial fulfilment of the requirement for the subject of ITL801 Blockchain Lab in degree of **Information Technology**, during the academic year **2024-2025** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Mr. Mandar Ganjapurkar
Guide

External Examiner(s)

- 1.
- 2.

Place: A.P. Shah Institute of Technology, Thane

Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Mr. Mandar Ganjapurkar**. Expressing gratitude towards our HoD, **Dr. Kiran Deshpande**, and the Department of Information Technology for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our peers for their helpful suggestions.

ABSTRACT

The notarization of documents plays a crucial role in establishing authenticity, ownership, and the integrity of information in various legal and official proceedings. However, traditional notary systems are often manual, time-consuming, and prone to manipulation, fraud, or loss of records. To overcome these limitations, this project presents a **Digital Notary System using Blockchain technology**, designed to provide a secure, decentralized, and transparent platform for document verification.

By utilizing the inherent properties of blockchain—immutability, decentralization, cryptographic security, and consensus mechanisms—the proposed system allows users to notarize documents digitally without the need for an intermediary. Once a document is uploaded and verified, its cryptographic hash is stored on the blockchain, ensuring that any tampering can be immediately detected. The use of **smart contracts** automates the verification and timestamping process, enhancing efficiency and reducing operational costs.

This system also enables stakeholders to verify the authenticity of documents in real-time, improving trust and reducing the risk of disputes. The solution is especially beneficial in sectors such as legal, educational, financial, and governmental institutions, where document integrity is paramount. By demonstrating the practical application of blockchain in notarization, this project aims to contribute to the evolution of secure and paperless digital ecosystems.

TABLE OF CONTENTS

1. Introduction...	1
1.1. Purpose.....	1
1.2. Problem Statement...	2
1.3. Objectives.	2
1.4. Scope.....	3
2. Literature Review...	4
3. Proposed System...	7
3.1. Features and Functionality...	7
4. Technical Specification.	9
5. Project Design.....	10
5.1. Use Case diagram.	10
5.2. DFD (Data Flow Diagram).....	11
5.3. System Architecture...	12
6. Results... ..	15
7. Conclusion... ..	17
8. Future Scope... ..	18
References	

1. Introduction

In an increasingly digital world, the demand for secure and efficient methods of verifying and authenticating documents has never been greater. Traditional notary services, while legally recognized, often involve physical presence, manual procedures, and reliance on centralized authorities. These systems are not only time-consuming but also susceptible to forgery, human error, and data loss. As the digital transformation of legal and administrative processes accelerates, there is a growing need for a modern solution that ensures transparency, trust, and integrity without the limitations of conventional methods.

Blockchain technology, known for its decentralized and tamper-proof nature, offers a revolutionary approach to solving these problems. By recording transactions and data on a distributed ledger that is immutable and transparent, blockchain ensures that once information is stored, it cannot be altered or deleted. This characteristic makes it an ideal foundation for building a secure Digital Notary System.

The aim of this project is to design and implement a Digital Notary System using blockchain to enable individuals and organizations to notarize documents in a decentralized, efficient, and legally verifiable manner. Through the use of cryptographic hashing and smart contracts, the system allows users to register, verify, and retrieve document proofs without the involvement of a third-party authority. The result is a trustless, auditable, and scalable notary platform that significantly reduces time, cost, and fraud.

This report outlines the motivation behind the project, the problems with existing systems, and how blockchain technology provides a feasible solution. It further explores the architecture, implementation, and real-world applications of the proposed system.

1.1 Purpose

The purpose of this project is to develop a Digital Notary System that utilizes Blockchain technology to replace traditional notarization methods with a more secure, transparent, and efficient digital solution. Traditional systems often require manual verification, physical presence, and rely on centralized authorities, making them time-consuming, costly, and prone to fraud or data tampering.

By leveraging blockchain's immutability and decentralization, the system allows users to notarize documents by generating cryptographic hashes and recording them on a distributed ledger along with timestamps and digital signatures. Smart contracts automate key processes, reducing the need for intermediaries and minimizing human error.

This project aims to improve document integrity, enhance user trust, and offer a scalable, cost-effective notarization solution for use across various sectors including legal, academic, corporate, and governmental environments. Ultimately, it showcases the practical potential of blockchain in building a modern, reliable, and tamper-proof notary system.

1.2 Problem Statement

Traditional notary systems are heavily reliant on manual processes and centralized authorities, making them slow, costly, and vulnerable to errors, fraud, and document tampering. The requirement for physical presence, paper-based documentation, and human verification creates inefficiencies and limits accessibility, especially in remote or digitally progressive environments. Moreover, once a document is notarized, it is often stored in centralized repositories that are susceptible to unauthorized access, data loss, and manipulation.

In a digital world where secure and verifiable information exchange is crucial, there is a clear need for a more robust, transparent, and tamper-resistant system for notarizing documents. The lack of automation and reliance on trust-based systems also poses challenges in verifying the authenticity of documents without involving third parties. Therefore, there is a pressing need to adopt a decentralized, automated approach that ensures integrity, transparency, and accessibility without compromising security.

This project addresses these issues by proposing a Blockchain-based Digital Notary System that eliminates central points of failure, provides immutable records, and automates document verification using smart contracts—ensuring a secure, efficient, and scalable notarization process.

1.3 Objectives

- To develop a decentralized digital notary system using blockchain technology.
- To eliminate the need for manual, paper-based, and centralized notarization processes.
- To ensure document integrity and authenticity through hashing and timestamping.
- To implement smart contracts for automating verification and notarization workflows.
- To provide a tamper-proof, transparent, and immutable ledger for notarized documents.
- To enhance trust among users by offering verifiable and auditable document records.
- To reduce the cost, time, and human intervention involved in traditional notary systems.
- To enable secure access and verification of documents from any location at any time.
- To demonstrate the practical application of blockchain in legal and administrative domains.

1.4 Scope

- Can provide a decentralized platform for notarizing digital documents securely.
- Can verify the authenticity of documents using blockchain-based cryptographic hashes.
- Can store immutable and time-stamped records on the blockchain for future reference.
- Can automate the notary process using smart contracts, reducing the need for intermediaries.
- Can support various types of documents, including legal, academic, and official records.
- Can reduce operational costs and time associated with traditional notarization processes.
- Can improve transparency and trust among stakeholders by offering publicly verifiable proofs.

2 Literature Review

Sr. No.	Title	Publish Year	Author(s)	Summary
1	<i>Blockchain Technology: Beyond Bitcoin</i>	2016	Crosby et al.	This foundational paper explores blockchain's potential beyond cryptocurrency, emphasizing its capabilities for ensuring data immutability and transparency across areas like digital identity and record verification.
2	<i>Decentralizing Privacy: Using Blockchain to Protect Personal Data</i>	2015	Zyskind et al.	The authors propose a decentralized personal data management framework using blockchain, allowing secure sharing and verification of personal data without centralized control—directly applicable to digital notary services.
3	<i>Smart Contract-Based Access Control for the Internet of Things</i>	2018	Zhang et al.	This study introduces smart contracts to enforce and automate access control policies in IoT environments, showcasing how similar automation can be leveraged in blockchain-based notarization systems.

3 Purposed System

The proposed system is a Blockchain-based Digital Notary Platform that enables secure, tamper-proof, and decentralized document verification and certification. This system aims to eliminate the need for physical presence, manual verification, and dependency on centralized authorities in traditional notary services. By leveraging blockchain technology, it ensures that once a document is notarized, it becomes immutable and publicly verifiable.

The platform allows users to register and upload documents that need notarization. When a document is uploaded, a unique cryptographic hash is generated using SHA-256 or a similar algorithm. This hash acts as a digital fingerprint of the document and is stored permanently on a blockchain ledger along with a timestamp and the user's digital identity. The actual document is not stored on the blockchain to maintain privacy—only its hash is saved, ensuring proof of existence and integrity.

Smart contracts are used to automate the notarization process. These contracts handle the logic for verifying, timestamping, and recording document hashes without requiring human intervention. In the case of multi-party notarization, all parties must digitally sign the document hash before it is committed to the blockchain.

The system can be built on public blockchains like Ethereum or private blockchains like Hyperledger depending on use-case and scalability needs. The front-end will be web-based for ease of access, while blockchain interaction will be managed using smart contract APIs and Web3 frameworks.

This proposed solution not only enhances the security and efficiency of notarization but also lays the foundation for a future-ready, legally compliant digital documentation ecosystem.

3.3 Features and Functionality

- **User Registration and Login –**
 - Secure registration and authentication system.
 - Each user has a unique digital identity linked with their notary activity.
- **Document Upload and Hashing –**
 - Users can upload any type of document (PDF, DOCX, etc.).
 - A cryptographic hash (e.g., SHA-256) is generated to represent the document's digital fingerprint.
- **Blockchain-based Notarization –**
 - Document hash along with timestamp and user ID is stored on the blockchain.
 - Guarantees immutability and proof of existence.
- **Smart Contract Automation –**
 - Smart contracts automatically handle document verification and storage of hash.

- **Document Integrity Verification –**
 - Users can re-upload a document to check whether it has been altered.
 - The system re-generates the hash and compares it with the one stored on the blockchain.
- **Timestamping and Audit Trail –**
 - All notarization events are timestamped.
 - Maintains a full history of document submissions and verifications.
- **Search and Verification Portal –**
 - Publicly accessible interface to verify if a document has been notarized.
 - Useful for third-party verifiers like legal bodies, banks, or employers.
- **Data Privacy –**
 - The actual document is never stored on the blockchain.
 - Only the hash is recorded, ensuring document confidentiality.
- **User Dashboard –**
 - Each user has access to a personal dashboard to manage their uploaded documents, view verification status, and check history.

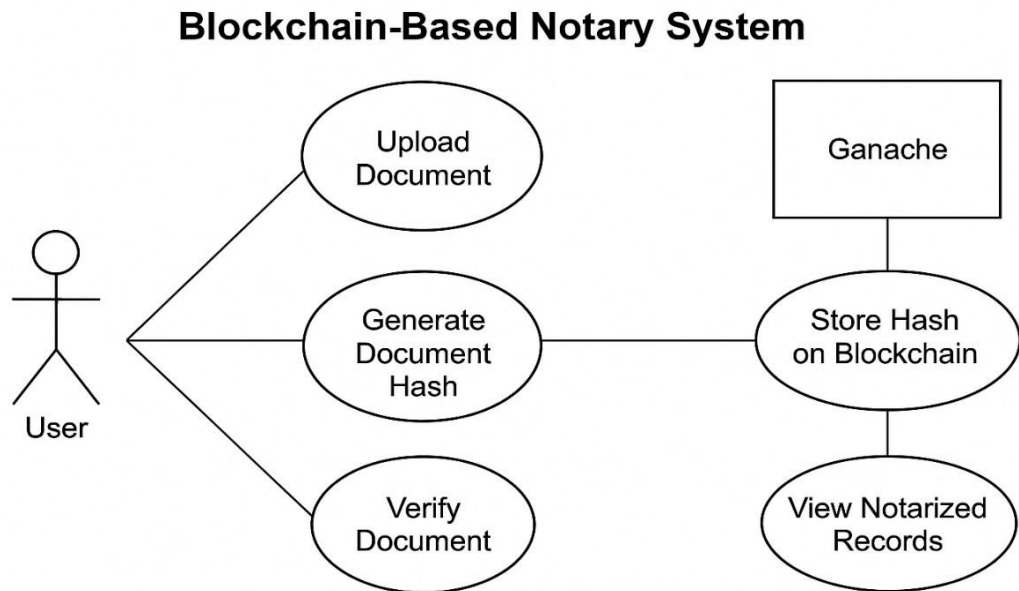
4 Technical Specification

The technology stack used in this project includes:

- Blockchain: Hardhat, Solidity 0.8.20, Ethers.js
- Frontend: HTML/CSS/JavaScript with Web3.js
- Development: Node.js, npm, Ganache

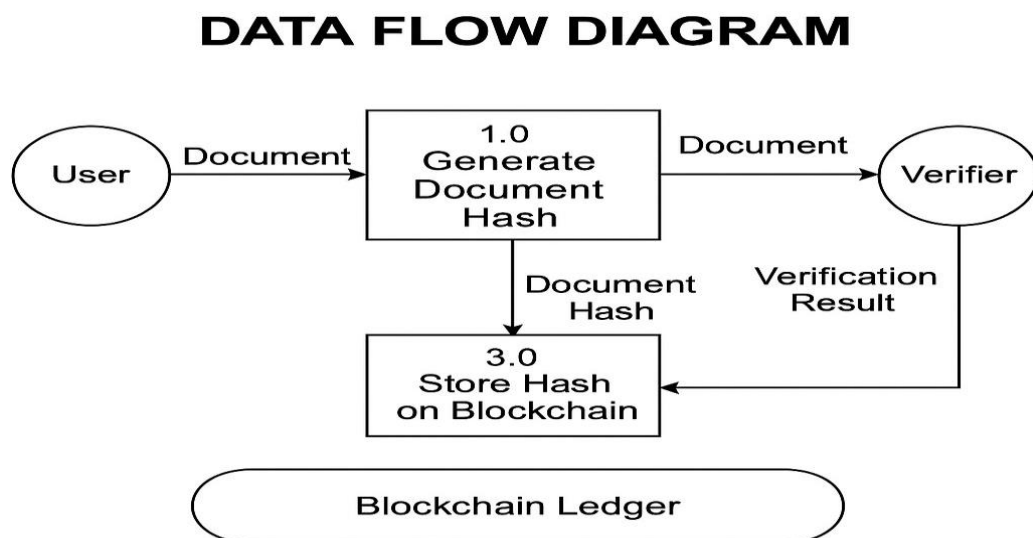
5 Project Design

5.1 Use Case Diagram



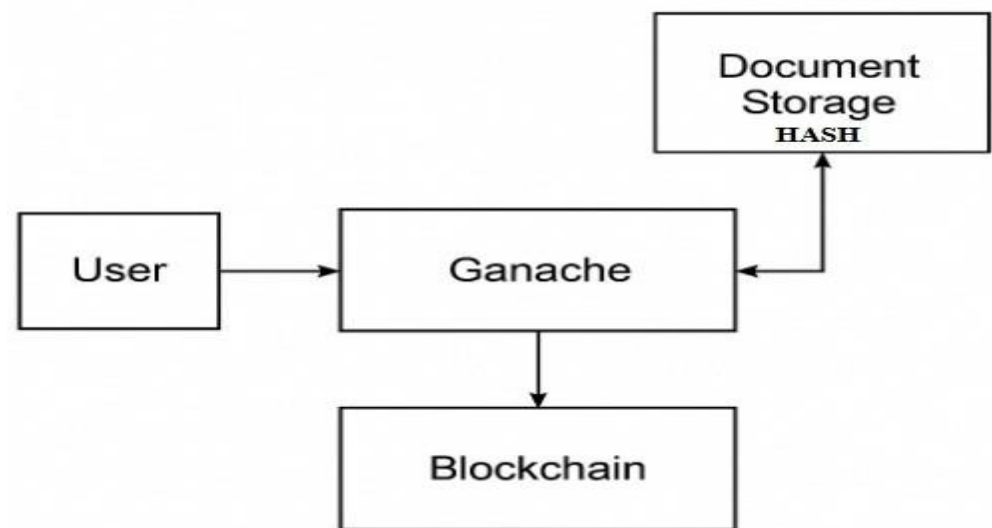
The Use Case Diagram outlines the primary actions users can perform, such as uploading documents, generating hashes, storing them on the blockchain, and verifying their authenticity, with the user interacting directly with the system's smart contract functionality.

5.2 DFD(Data Flow Diagram)



The Data Flow Diagram (DFD) captures the logical flow of data through the system, showing how documents are processed into hashes, stored on the blockchain, and later retrieved for verification, clearly defining the interactions between users, processes, and data stores in a structured manner according to software engineering standards.

5.3 System Architecture



The System Architecture Diagram illustrates the core components of the blockchain-based notary system, including the user interface, hashing module, Web3-based interaction with the Ganache blockchain, and optional document storage.

6 Results

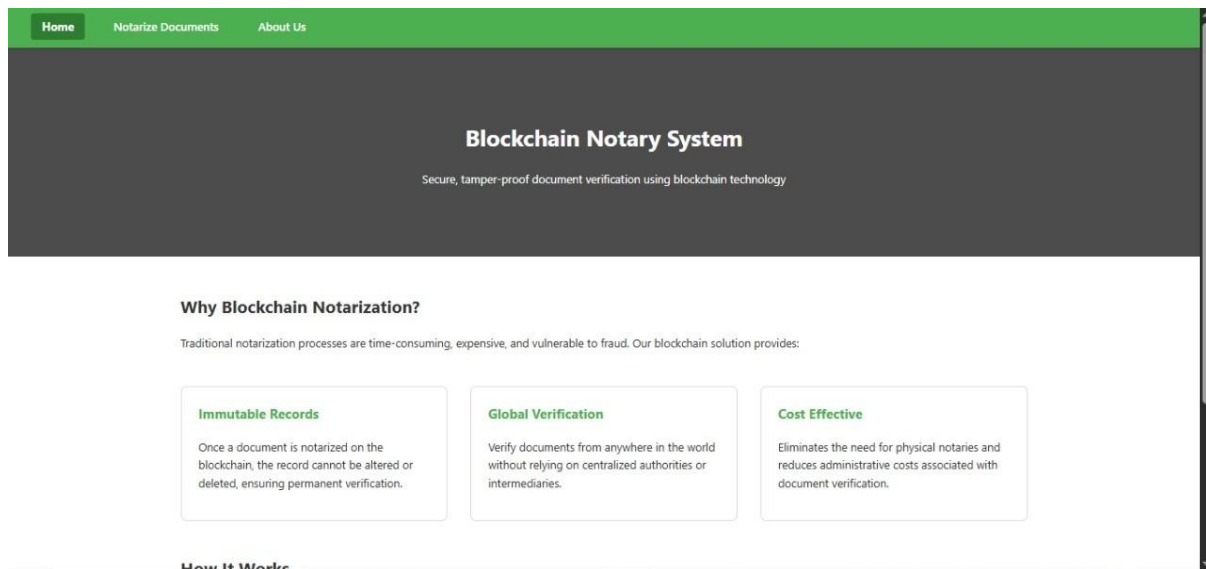


Fig 6.1

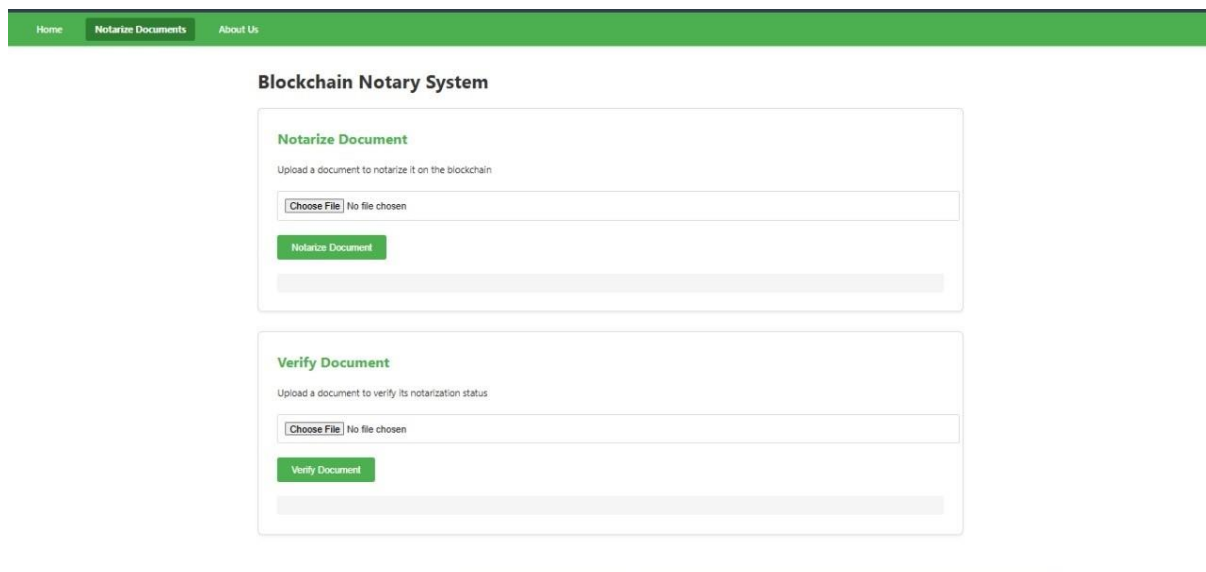


Fig 6.2

The successful implementation of the Digital Notary System using blockchain demonstrates the practicality and effectiveness of decentralization in modern document verification processes. The developed platform allows users to upload digital documents, which are then processed to generate unique cryptographic hashes. These hashes are securely stored on a blockchain along with timestamps, ensuring that every document is recorded in an immutable and traceable manner. This process guarantees that once a document is notarized, any alteration or forgery attempt can be easily detected by comparing the current hash with the original one stored on the blockchain.

Additionally, the integration of smart contracts within the system automates the verification and validation process, removing the need for third-party intermediaries. This not only streamlines the workflow but also significantly reduces the possibility of human errors and manipulation. The results show that the system operates efficiently, maintaining data integrity, providing proof of authenticity, and enhancing transparency. The platform supports quick access to verification data and can serve as a tamper-proof record, proving its potential as a viable alternative to traditional notarization methods.

7 Conclusion

This project effectively addresses the limitations of traditional notary systems by introducing a blockchain-based digital solution that emphasizes security, transparency, efficiency, and decentralization. Traditional notarization involves manual paperwork, physical presence, and dependency on centralized authorities, all of which make the process slow, costly, and vulnerable to manipulation or loss. By adopting blockchain technology, the proposed system eliminates these drawbacks and presents a more robust and scalable framework for notarizing documents.

Through the use of cryptographic hashing, time-stamping, and automated smart contracts, the Digital Notary System ensures that every document is verifiably authentic and permanently recorded. It offers a high level of trust without requiring users to rely on any single authority. This approach paves the way for a future where legal, academic, and business documentation can be handled with enhanced reliability and speed. Overall, the project not only proves the technical feasibility of such a system but also lays the foundation for a secure, paperless, and efficient notarization process in the digital era.

8 Future Scope

While the current implementation achieves the core goal of decentralized digital notarization, there is significant potential to expand and improve the system in the future. One major enhancement could be the integration of user identity verification through national ID systems like Aadhaar (India) or other biometric authentication methods. This would ensure that only verified individuals can notarize or access sensitive documents, thereby increasing the legal validity and trust in the system. Additionally, future versions can include role-based access control, allowing specific levels of document access depending on user roles (e.g., client, authority, verifier).

The system can also be extended to handle multi-party notarization, where multiple stakeholders can digitally sign and verify a single document. Support for cross-chain interoperability and integration with private or consortium blockchains can be explored to suit the varying needs of different organizations and government bodies. Furthermore, developing mobile and offline access features would make the platform more accessible to users in remote areas or those with limited internet connectivity. Legal recognition and compliance with international electronic notary standards and digital signature laws will be crucial in enabling wide-scale adoption of this technology across jurisdictions and industries.

References

1. Crosby, M., Pattanayak, P., Verma, S., & Kalyanaraman, V. (2016). Blockchain Technology: Beyond Bitcoin. *Applied Innovation Review*.
2. Zyskind, G., Nathan, O., & Pentland, A. (2015). Decentralizing Privacy: Using Blockchain to Protect Personal Data. *IEEE Security and Privacy Workshops*.
3. Zhang, Y., Kasahara, S., Shen, Y., Jiang, X., & Wan, J. (2018). Smart Contract-Based Access Control for the Internet of Things. *IEEE Internet of Things Journal*.
4. Pilkington, M. (2016). Blockchain Technology: Principles and Applications. In *Research Handbook on Digital Transformations*.
5. Aitzhan, N. Z., & Svetinovic, D. (2018). Security and Privacy in Decentralized Energy Trading through Multi-Signatures, Blockchain and Anonymous Messaging Streams. *IEEE Transactions on Dependable and Secure Computing*.
6. Underwood, S. (2016). Blockchain Beyond Bitcoin. *Communications of the ACM*, 59(11), 15-17.
7. Yli-Huumo, J., Ko, D., Choi, S., Park, S., & Smolander, K. (2016). Where Is Current Research on Blockchain Technology?—A Systematic Review. *PloS One*, 11(10), e0163477.
8. Kshetri, N. (2017). Can Blockchain Strengthen the Internet of Things? *IT Professional*, 19(4), 68-72.
9. Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System. [White Paper].
10. Tapscott, D., & Tapscott, A. (2016). *Blockchain Revolution: How the Technology Behind Bitcoin is Changing Money, Business, and the World*. Penguin.